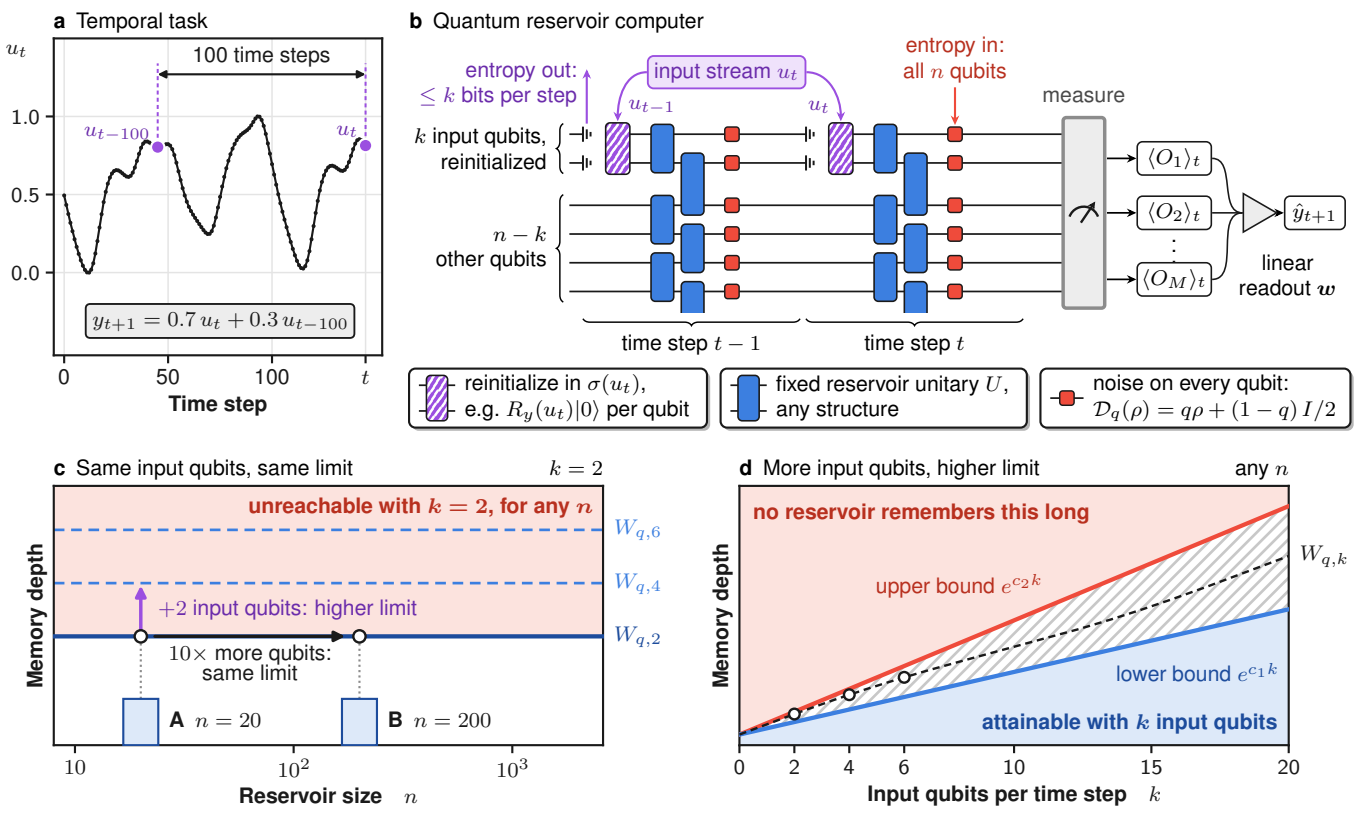

a Temporal task
100 time steps
u_t
u_{t-100}
$y_{t+1} = 0.7\,u_t + 0.3\,u_{t-100}$
Time step
b Quantum reservoir computer
entropy out: $\leq k$ bits per step
input stream u_t
entropy in: all n qubits
measure
k input qubits, reinitialized
$n-k$ other qubits
$\langle O_1\rangle_t$
$\langle O_2\rangle_t$
$\langle O_M\rangle_t$
$\hat{y}_{t+1}$
linear readout $\boldsymbol{w}$
time step $t-1$
time step t
reinitialize in $\sigma(u_t)$, e.g. $R_y(u_t)|0\rangle$ per qubit
fixed reservoir unitary U, any structure
noise on every qubit: $\mathcal{D}_q(\rho) = q\rho + (1-q)\,I/2$
c Same input qubits, same limit $k=2$
unreachable with $k=2$, for any n
$W_{q,6}$
$W_{q,4}$
$W_{q,2}$
+2 input qubits: higher limit
10× more qubits: same limit
A $n=20$
B $n=200$
Memory depth
Reservoir size n
d More input qubits, higher limit any n
no reservoir remembers this long
upper bound $e^{c_2 k}$
$W_{q,k}$
lower bound $e^{c_1 k}$
attainable with k input qubits
Memory depth
Input qubits per time step k